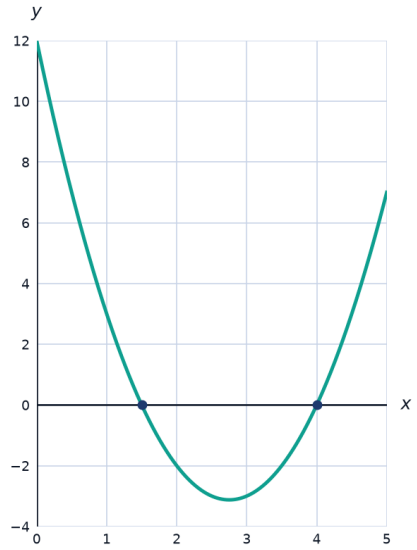
Quadratics & Nonlinear Equations

Section 1: Factoring & Zero Product Property

1 What is the sum of the solutions to the quadratic equation $2x^2 - 11x + 12 = 0$?



a. $\frac{11}{2}$

b. $\frac{12}{2}$

c. $-\frac{11}{2}$

d. 6

2 If $(2x - 3)(x + 5) = 0$, what is the positive solution to the equation?

a. $\frac{2}{3}$

b. $\frac{3}{2}$

c. 5

d. -5

3 If $x^2 - 16 = (x - c)(x + c)$ for all real numbers x, where $c > 0$, what is the value of c?

a. 2

b. 4

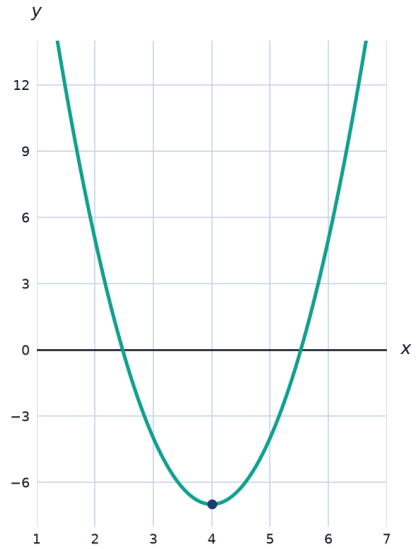
c. 8

d. 16

Section 2: Vertex Form & Minima/Maxima

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4 A parabola in the xy-plane has equation $y = 3(x - 4)^2 - 7$. What are the coordinates of its vertex?



a. $(-4, -7)$

b. $(4, -7)$

c. $(4, 7)$

d. $(-4, 7)$

5 What is the minimum value of the quadratic function $f(x) = x^2 - 6x + 14$?

a. 3

b. 5

c. 9

d. 14

6 The height h in feet of a projectile t seconds after launch is given by $h(t) = -16t^2 + 64t + 80$. What is the maximum height, in feet, reached by the projectile?

a. 120

b. 144

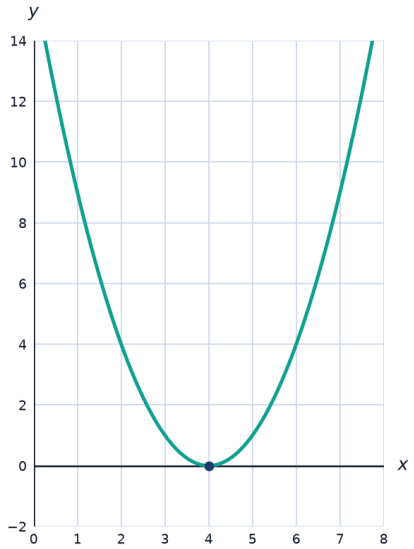
c. 160

d. 184

Section 3: Discriminant Analysis ($b^2 - 4ac$)

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7 For what value of c does the equation $x^2 - 8x + c = 0$ have exactly one real solution?



a. 8

b. 16

c. 32

d. 64

8 Which of the following quadratic equations has no real solutions?

a. $x^2 - 4x + 3 = 0$

b. $x^2 - 2x + 1 = 0$

c. $x^2 - 2x + 5 = 0$

d. $x^2 + 3x - 4 = 0$

9 If the quadratic equation $3x^2 - kx + 12 = 0$ has two distinct real roots, which condition must k satisfy?

a. $k^2 > 144$

b. $k^2 < 144$

c. $k^2 = 144$

d. $k > 12$

Section 4: Nonlinear Systems & Intersections

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10 How many points of intersection exist between the line $y = 2x - 3$ and the parabola $y = x^2 - 4x + 6$?

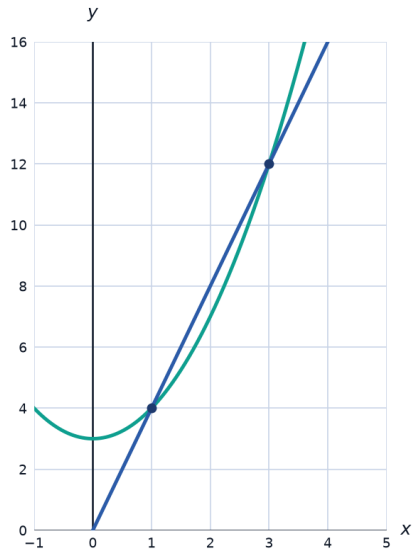
a. 0

b. 1

c. 2

d. 3

11 If (x, y) is a solution to the system $y = x^2 + 3$ and $y = 4x$, what is the sum of all possible values of x?



a. 3

b. 4

c. 7

d. 12

12 A circle in the xy-plane has equation $(x - 3)^2 + (y + 2)^2 = 25$. At what points does the line $x = 3$ intersect the circle?

a. $(3, 3)$ and $(3, -7)$

b. $(3, 5)$ and $(3, -5)$

c. $(0, 3)$ and $(0, -7)$

d. $(3, 2)$ and $(3, -2)$

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